Antiderivatives and Riemann Sums

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic antiderivatives

Find the general antiderivative of each function:

a $f(x) = x^4$

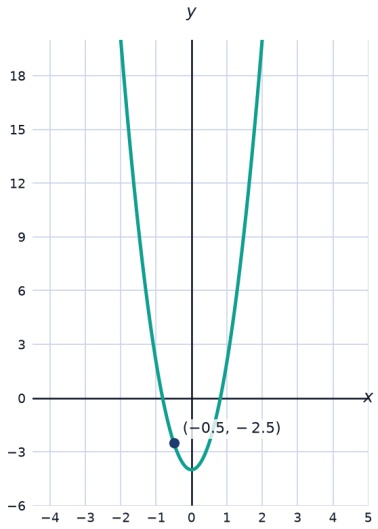
b $f(x) = \frac{1}{x}$

c $f(x) = e^x$

d $f(x) = \sec^2 x$

2 [Foundational] • Target: Basic antiderivatives

Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.



3 [Foundational] • Target: Riemann sums: left, right, and midpoint

Approximate $\int_0^4 x^2\,dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.

4 [Foundational] • Target: Riemann sums: left, right, and midpoint

Approximate $\int_0^4 x^2\,dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

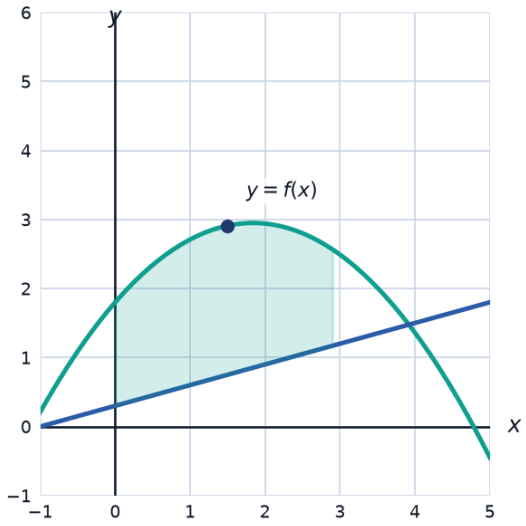
5 [Foundational] • Target: Riemann sums: left, right, and midpoint

Approximate $\int_0^4 x^2\,dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.

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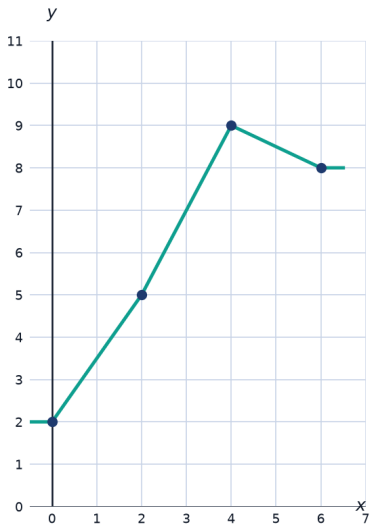
6 [Foundational] • Target: The Trapezoidal Rule

Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2\,dx$.



7 [Foundational] • Target: The Trapezoidal Rule

A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x)\,dx$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums as limits; the definite integral

Explain how the definite integral $\int_a^b f(x)\,dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.

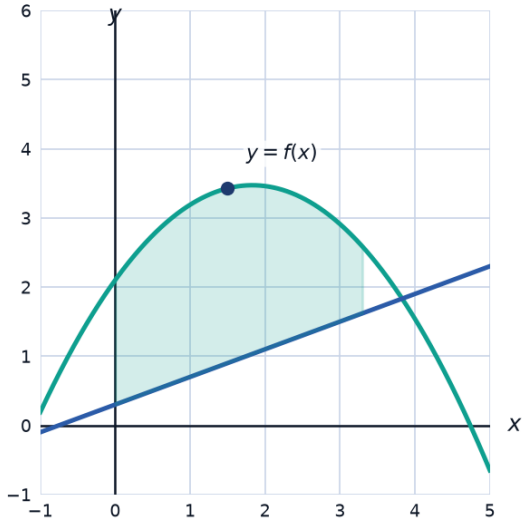
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9 [Intermediate] • Target: Sums as limits: the definite integral

Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2\,dx$, and explain why using the shape of the graph.

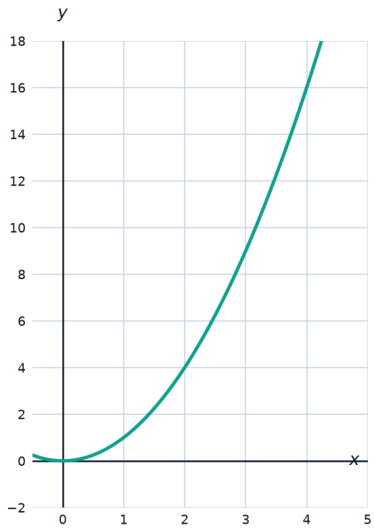
10 [Intermediate] • Target: Riemann sums from a table of values

A table gives $f(0) = 3$, $f(1) = 5$, $f(2) = 8$, $f(3) = 10$, $f(4) = 9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.



11 [Intermediate] • Target: Visualizing a Riemann sum approximation

The graph shows $f(x) = x^2$ on $[0, 4]$. Use the graph to explain why the Right Riemann Sum $R_4 = 30$ (found earlier) overestimates the true area, and why the Left Riemann Sum $L_4 = 14$ underestimates it.



12 [Intermediate] • Target: An applied antiderivative problem

A factory's production rate is $r(t) = 100 + 20t$ units/day, where t is days since a process change. Find the total production function $P(t)$ given $P(0) = 0$, then find the total production over the first 10 days.

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13 [Intermediate] • Target: More antiderivative practice

Find the general antiderivative of $f(x) = 4x^3 - \frac{2}{x^2} + \cos x$.

14 [Intermediate] • Target: Comparing Riemann sums to the exact integral

For $\int_0^4 x^2\,dx$, the exact value is $\frac{64}{3} \approx 21.33$. Rank the four approximations found earlier — $L_4 = 14$, $R_4 = 30$, $M_4 = 21$, $T_4 = 22$ — from closest to farthest from the exact value, and explain which type of approximation (midpoint or trapezoidal) tends to be more accurate for a smooth curving function.

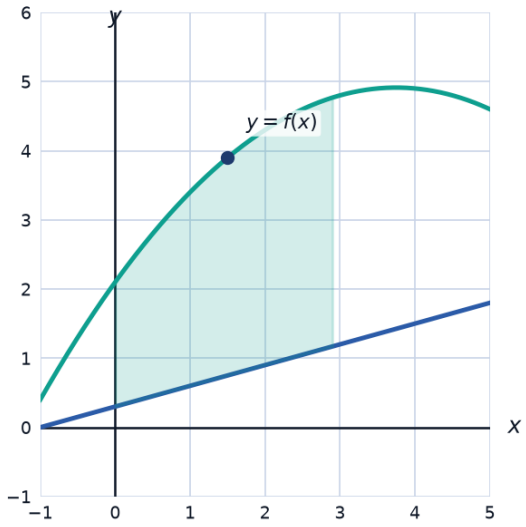
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Level 1 · Foundations Find the exact sum of the infinite geometric series: $\sum_{n=1}^{\infty} 3\left(-\frac{2}{5}\right)^n$

16 [Challenging] • Target: Antiderivatives of trigonometric and exponential functions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Level 1 · FoundationsExpress the infinite summation limit as a definite integral on the interval $[0, 1]$ and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{i^3}{n^4}$.



17 [Challenging] • Target: Visualizing a Riemann sum approximation • Cross-Lesson: 8.01 Area Between Curves

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 [Challenging] • Target: An applied antiderivative problem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Level 2 · Medium Evaluate the indefinite integral using partial fraction decomposition: $\int \frac{5x - 3}{x^2 - 2x - 3}\,dx$

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19 [Challenging] • Target: More antiderivative practice • Cross-Lesson: 1.01 Limits Graphically and Numerically

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.

20 [Challenging] • Target: Comparing Riemann sums to the exact integral • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Does $\sum_{n=0}^{\infty} (0.9)^n$ converge, and to what?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic antiderivatives

Find C in the particular antiderivative $F(x) = x^3 - 7x + C$ of $f(x) = 3x^2 - 7$, satisfying $F(-1) = -8$.

a. -8

b. -12

c. -14

d. -16

22 [Foundational] • Target: Riemann sums: left, right, and midpoint

Approximate $\int_0^6 f(x)\,dx$ using the Trapezoidal Rule (equal width 2), given $f(0) = 7$, $f(2) = 12$, $f(4) = 9$, $f(6) = 3$.

a. 54

b. 52

c. 62

d. 50

23 [Foundational] • Target: The Trapezoidal Rule

For $f(x) = 4x^2$ on $[0, 5]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^5 f(x)\,dx$.

a. Both Left and Right Riemann Sums underestimate.

b. Both Left and Right Riemann Sums overestimate.

c. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.

d. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

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24 [Foundational] • Target: Sums as limits: the definite integral

A factory's production rate is $r(t) = 80 + 26t$ units/day, where t is days since a process change. Find the total production $P(2)$ given $P(0) = 0$.

a. 160

b. 238

c. 186

d. 212

25 [Foundational] • Target: Riemann sums from a table of values

For $f \geq 0$ continuous on $[a, b]$, FTC connects $\int_a^b f(x)dx$ to the area under the curve because Riemann sums $\sum f(x_i^*)\Delta x$ converge to the integral as:

a. $n \to \infty$ (partition width $\to 0$)

b. $b \to \infty$

c. $f \to 0$

d. $x \to a$

26 [Foundational] • Target: Antiderivatives of trigonometric and exponential functions

Write $z = 4(\cos 180^\circ + i\sin 180^\circ)$ in the form $x + yi$ (exact values).

a. $-4 + 1$

b. 4

c. -4

d. $-4i$

27 [Foundational] • Target: Visualizing a Riemann sum approximation

Consider the function $f(x) = \frac{\sin(2x)}{\cos(x)}$. For the domain $x \in [0, 2\pi]$ such that $\cos(x) eq 0$, find the exact value of the definite integral $\int_0^{\pi/2} f(x)dx$.

a. $\ln(2)$

b. 1

c. 2

d. $\ln(1)$

28 [Foundational] • Target: An applied antiderivative problem

Which is an equation of the plane through $(-3,\ 1,\ 4)$, $(-2,\ 2,\ 4)$ and $(1,\ -1,\ 1)$?

a. $x - y + 2z = -4$

b. $x + y + 2z = 6$

c. $x - y + 2z = 4$

d. $x + y = -2$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: More antiderivative practice**

Find the arc length of $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}\,t \rangle$ for $0 \le t \le \ln(2)$.

a. $\frac{3}{2}$

b. 3

c. $\frac{3}{4}$

d. $\frac{5}{2}$

30 **[Intermediate]** • **Target: Comparing Riemann sums to the exact integral**

Find the area of the triangle with vertices $(1,\ 0,\ 2), (1,\ 1,\ 2), (-3,\ 3,\ -3)$.

a. $\sqrt{41}$

b. $\frac{\sqrt{41}}{3}$

c. $\frac{\sqrt{41}}{2}$

d. $\frac{3}{2}$

31 **[Intermediate]** • **Target: Basic antiderivatives**

Apply the ratio test to $\sum_{n=1}^{\infty} \frac{6^n}{n!}$. What is $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$?

a. 0

b. ∞

c. 1

d. 6

32 **[Intermediate]** • **Target: Riemann sums: left, right, and midpoint**

Gradient vector $ablaf = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$. For $f = x^2 + y^2$ at $(1, 1, 0)$:

a. $(1, 1, 0)$

b. $(1, 2, 0)$

c. $(0, 0, 0)$

d. $(2, 2, 0)$

33 **[Intermediate]** • **Target: The Trapezoidal Rule**

Continuity of piecewise: $f(x) = \{kx, x < 2; x^2 - 2, x \ge 2$ is continuous iff $k =$

a. 1

b. 2

c. 0

d. 3

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34 [Intermediate] • Target: Sums as limits: the definite integral

Let $f(x) = x^3 + 3x - 6$, which is one-to-one. Find $(f^{-1})'(-6)$.

a. $\frac{1}{3}$

b. 3

c. $\frac{1}{111}$

d. $-\frac{1}{6}$

35 [Intermediate] • Target: Riemann sums from a table of values

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$.

a. 0

b. 2

c. ∞

d. $\frac{1}{2}$

36 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find the distance from $(2,\ -2,\ 0)$ to the plane $-3y + z = -4$.

a. 10

b. $\frac{\sqrt{10}}{5}$

c. 1

d. $\sqrt{10}$

37 [Intermediate] • Target: Visualizing a Riemann sum approximation

For the Dirichlet function $f(x) = \{1, x \in \mathbb{Q}; 0, x \not\in \mathbb{Q}$, $\lim_{x \to 0} f(x)$:

a. equals 0

b. equals 1

c. equals $\frac{1}{2}$

d. does not exist

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: An applied antiderivative problem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Evaluate $\tan^{-1}(1)$ in degrees.

a. 67°

b. 22°

c. 69°

d. 45°

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39 [Challenging] • Target: More antiderivative practice • Cross-Lesson: 1.01 Limits Graphically and Numerically

Telescoping series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right) =$

a. ∞

b. 1

c. 0

d. 1/2

40 [Challenging] • Target: Comparing Riemann sums to the exact integral • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Bessel function $J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2}\left(\frac{x}{2}\right)^{2m}$. Radius of convergence:

a. 1

b. ∞

c. π

d. 2

41 [Challenging] • Target: Basic antiderivatives • Cross-Lesson: 2.01 The Definition of the Derivative

For iterated integral $\int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)dy \right) dx$: the inner integral is done first, then the outer. True or false?

a. False — outer first

b. True

c. Only for separable functions

d. Depends on Fubini

42 [Challenging] • Target: Riemann sums: left, right, and midpoint • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R = 3$ and the center of the series is $a = 2$, which of the following intervals represents the values of x for which the series converges?



a. $(-1, 5)$

b. $[-1, 5]$

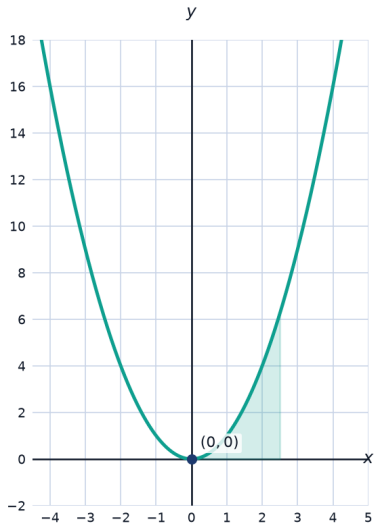
c. $(-1, 5]$

d. $[-1, 5)$

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43 **[Challenging]** • **Target: The Trapezoidal Rule** • **Cross-Lesson: 3.01 The Chain Rule**

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.



a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_{0}^{4}((4 - x^2)^2 - r^2)dx$

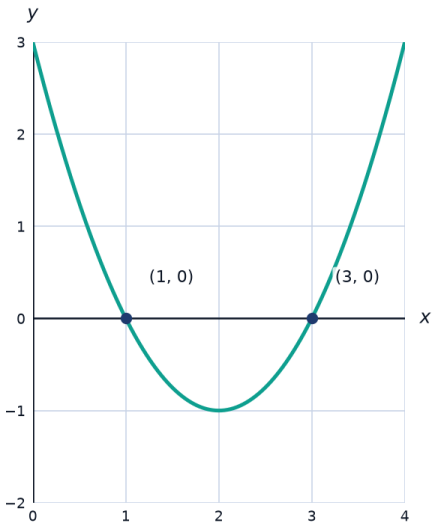
c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_{1}^{4}((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

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44 [Challenging] • Target: Sums as limits: the definite integral • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. What is the definite integral of $f(x)$ from $x = 1$ to $x = 3$?



a. $-\frac{4}{3}$

b. 0

c. $\frac{4}{3}$

d. 2

45 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Consider the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n \cdot 3^n}$. For what values of x does this series converge conditionally?

a. $-1 \leq x < 2$

b. $-1 < x \leq 2$

c. $-1 \leq x \leq 2$

d. $-1 < x < 2$

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